

Week 6 Problem Set

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1. Start a new document, problemset6.Rmd, in your soc10problemsets repository. (You can download a template here, but be sure to save it in your own soc10problemsets repo). Create a network based on 5 people in our class! (You can use the “class_activities.csv” file from Canvas if helpful or use the code in the template). Using **igraph**, create a network diagram based on who has worked with who in our class activities this quarter (you may want to use **graph_from_data_frame()** if you read in the csv). Then plot the network. (Note: a network with only 5 people, excluding ties to everyone outside of these 5, is sufficient). **Commit** your changes.

```
library(readr)
library(RCurl)
```

```
## Warning: package 'RCurl' was built under R version 4.5.2
```

```
library(magrittr)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(igraph)
```

```
## Warning: package 'igraph' was built under R version 4.5.2
```

```
##
## Attaching package: 'igraph'
```

```
## The following objects are masked from 'package:dplyr':
##
##   as_data_frame, groups, union
```

```

## The following objects are masked from 'package:stats':
##
##   decompose, spectrum

## The following object is masked from 'package:base':
##
##   union

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/class_activities.csv")

# download class activities data (from github)
class_activities <- read_csv(url)

## Rows: 71 Columns: 3

## -- Column specification -----
## Delimiter: ","
## chr (2): name_1, name_2
## dbl (1): n
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

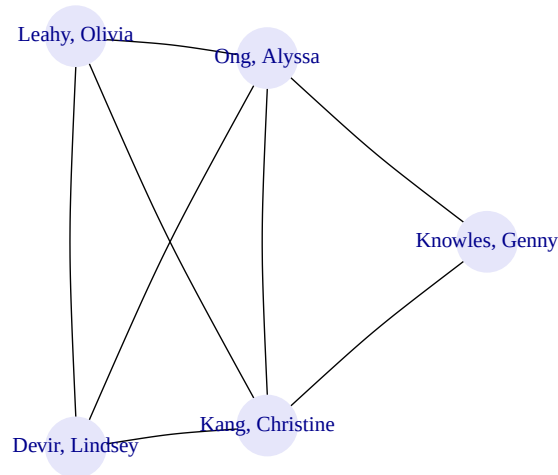
five_names <- c("Devir, Lindsey", "Leahy, Olivia", "Kang, Christine",
               "Ong, Alyssa", "Knowles, Genny")

five_people <- class_activities %>%
  filter(name_1 %in% five_names & name_2 %in% five_names)

# network
net <- graph_from_data_frame(five_people, directed = FALSE)

# plot
plot(net, vertex.size = 30,
      vertex.color = "lavender",
      vertex.frame.color = NA,
      vertex.label.cex = .7,
      edge.curved = .05,
      edge.arrow.size = .3,
      edge.width = .7,
      edge.color = "gray1")

```



2. Choose a centrality measure, and explain how you would interpret this in the context of the class activities data. Who is/are the most central actor(s) according to this measure? **Commit** your changes.

```
degree(net)
```

```
## Devir, Lindsey Kang, Christine Knowles, Genny Leahy, Olivia Ong, Alyssa
##          3          4          2          3          4
```

I chose degree centrality as my measure and to define from class, degree centrality counts how many edges each node has, so the node with the most edges is most central. In the context of class activities from my plotting, degree centrality identifies who has worked with the most classmates which captures the most active collaborator to this specific group of students and assigned to work with the most amount of students in this specific group of 5 students. Using the `degree()` function, Christine and Alyssa are tied for the highest degree centrality, each with a score of 4, meaning they have worked with the most other students (a total of 4) in the group; therefore the most central actors based on degree centrality.

3. Now download the IPCC co-authorship data (you can use the code included in the template). Using the IPCC co-authorship data, create a network graph of all co-authors who share at least 2 publications together, excluding those with no network ties, similar to what we do in section @ref(co-authorship-patterns). Graph the data using eigenvector scores, rather than betweenness scores, to plot the vertex sizes. Does this make a difference in our interpretation of the network? **Commit** your changes.

Using eigenvector centrality rather than betweenness centrality shifts which authors appear most central because while betweenness identifies authors who bridge different parts of the network, eigenvector centrality

rewards authors connected to other well-connected authors meaning an author with fewer total collaborations can still appear large if their collaborators are themselves highly connected. Looking at the network, authors like Nakicenovic, Riahi, and Carraro appear among the most central nodes, so they are deeply embedded in the core cluster rather than simply their role as bridges in the cluster. This shifts our interpretation of the authors toward a tightly knit inner circle of researchers who repeatedly collaborate with each other. This has implications for IPCC authorship because it suggests that influence in this network is concentrated among a small group of highly interconnected scholars, which may reinforce existing geographic and institutional biases in whose perspectives shape global climate policy if only certain regions and beliefs are represented.

```
library(readr)
library(RCurl)
library(magrittr)
library(dplyr)
library(igraph)

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_co_authorship.csv")

# download authorship data (from github)
ipcc_authors <- read_csv(url, skip = 1)

# replace NA's with 0's
ipcc_authors %<>%
  mutate_all(~replace(., is.na(.), 0))

# reorder column names to match rows
ipcc_authors %<>%
  select(order(colnames(.)))

# turn dataframe into matrix
ipcc_authors %<>%
  select(-Author) %>%
  as.matrix()

# replace 1s with 0s
ipcc_authors[ipcc_authors == 1] = 0

# replace NAs with 0s
ipcc_authors[is.na(ipcc_authors)] = 0

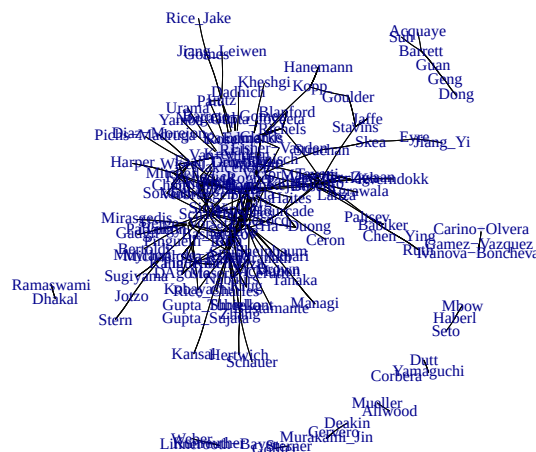
# vector of authors who do not collaborate with others
no_colab <- which(apply(ipcc_authors, 1, sum, na.rm = TRUE) == 0)

# remove non-collaborators
ipcc_authors <- ipcc_authors[-no_colab, -no_colab]

# create network
ipcc_net <- graph_from_adjacency_matrix(ipcc_authors,
                                         mode = "undirected")

## Warning: The 'adjmatrix' argument of 'graph_from_adjacency_matrix()' must be symmetric
## with mode = "undirected" as of igraph 1.6.0.
## i Use mode = "max" to achieve the original behavior.
## This warning is displayed once per session.
```

```
# plot network with eigenvector scores
plot(ipcc_net,
      vertex.size = V(ipcc_net)$eigenvector/max(V(ipcc_net)$eigenvector) * 20,
      vertex.color = "lavender",
      vertex.frame.color = NA,
      vertex.label.cex = .5,
      edge.curved = .05,
      edge.width = .1,
      edge.color = "gray1")
```



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To disseminate the memo most efficiently across the IPCC co-authorship network, I would give it to Nakicenovic, who has the highest betweenness centrality score among all 148 authors in the network. Betweenness centrality measures how often a given node lies on the shortest path between two other nodes, meaning that authors with high betweenness scores act as critical bridges connecting otherwise disconnected parts of the network. This makes betweenness the most appropriate centrality measure for this dissemination task, since we want someone who can pass information not just to their immediate collaborators, but across many different clusters of researchers. Since Nakicenovic sits at the intersection of multiple collaborative clusters, giving them the memo maximizes the likelihood that it reaches the broadest range of authors through their existing co-authorship connections. In contrast, choosing someone with high degree or eigenvector centrality might concentrate the memo within one densely connected cluster, missing large portions of the broader network.

```
# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_co_authorship.csv")

# download authorship data (from github)
ipcc_authors <- read_csv(url, skip = 1)

# replace NA's with 0's
ipcc_authors %<>%
  mutate_all(~replace(., is.na(.), 0))

# reorder column names to match rows
ipcc_authors %<>%
  select(order(colnames(.)))

# turn dataframe into matrix
ipcc_authors %<>%
  select(-Author) %>%
  as.matrix()

# replace 1s with 0s (threshold: at least 2 co-authorships)
ipcc_authors[ipcc_authors == 1] = 0

# replace NAs with 0s
ipcc_authors[is.na(ipcc_authors)] = 0

# vector of authors who do not collaborate with others
no_colab <- which(apply(ipcc_authors, 1, sum, na.rm = TRUE) == 0)

# remove non-collaborators
ipcc_authors <- ipcc_authors[-no_colab, -no_colab]

# network
ipcc_net <- graph_from_adjacency_matrix(ipcc_authors,
                                         mode = "undirected")

# authors in descending order of betweenness
V(ipcc_net)[order(betweenness(ipcc_net), decreasing = TRUE)]

## + 148/148 vertices, named, from a6c4edf:
##   [1] Nakicenovic      Smith             Shukla
##   [4] Price            Edenhofer        Schaeffer
##   [7] Krey             Bosetti          Carraro
```

```
## [10] Riahi           Hourcade           Uerge-Vorsatz
## [13] Rogner           Sathaye            Van-Vuuren
## [16] Clarke           Victor             Faaaj
## [19] Lutz             Michaelowa         Paltsev
## [22] Strachan         Edmonds            Den-Elzen
## [25] Hoehne           Zhang              Skea
## [28] Kopp             Stavins            Kahn-Ribeiro
## + ... omitted several vertices
```

5. Finally, plot the IPCC co-authorship data, but change the color/shape of the vertices so that these correspond to authors' gender. Describe what you see. What conclusions can you make about authorship and gender? (Note that you will need the IPCC_cv.csv dataset. The code in the template has combined these datasets). **Commit** your changes.

The network I created reveals a gender imbalance in IPCC co-authorship because male authors (blue) overwhelmingly dominate the network with the vast majority of both the central cluster and the nodes being blue. Female authors (red) are fewer in number and tend to appear more on the outside of the network, with only a handful visible near the core. This suggests that not only are women underrepresented in IPCC authorship overall, but those who are included tend to have fewer co-authorship connections than their male counterparts. However, this data only captures co-authorship patterns prior to the AR5 report, and it is possible that representation has improved in more recent assessment cycles and additionally, the binary coding of gender in this dataset does not capture the full spectrum of gender identities among researchers which is important to consider and understand about this data and when considering gender as a variable.

```
library(readr)
library(RCurl)
library(magrittr)
library(dplyr)
library(igraph)

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_cv.csv")

# download authorship data (from github)
ipcc_cvs <- read_csv(url)

# remove authors who did not collaborate
ipcc_cvs %<>%
  slice(-no_colab)

# define gender variable
ipcc_cvs %<>%
  mutate(gender_color = case_match(Gender,
                                    1 ~ "steelblue",
                                    0 ~ "tomato",
                                    .default = "gray"))

## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'gender_color = case_match(Gender, 1 ~ "steelblue", 0 ~
##   "tomato", .default = "gray")'.
## Caused by warning:
## ! 'case_match()' was deprecated in dplyr 1.2.0.
## i Please use 'recode_values()' instead.
```

